

Kloqra

Time analytics engine for agencies & product teams

Technical demo + 2026–2027 roadmap
June 2026

The problem we solve

Pain	Kloqra answer
Time trapped in spreadsheets	Timer + timesheet + time tracker — one ledger
"Did they submit?"	Per-project approval workflow + realtime notifications
Payroll / client reports	Export wizard, schedules, invoice PDF, public API
Admin blind spots	Dashboard, utilization, team live presence

Capture → approve → bill → export — without spreadsheet surgery.

Vision & north-star outcomes

Stakeholder	Success looks like
Member	Log time in <10s; know submission status; live UI on workflow events
Admin	Burn, utilization, approvals in one place; export without Excel
Agency owner	Multi-project profitability; client-ready reports
Engineering	Contract-first; CI gates; safe deploys; observable production

Guiding principles

- **Manual ledger before automation** — time logs + approvals are source of truth
- **Contract-first** — packages/contracts before API or UI
- **REST is truth, push is hint** — WebSocket invalidates; pages REST-refetch
- **Member privacy** — no org revenue or peer rankings on client
- **Small shippable epics** — one phase per PR

Architecture

```
apps/app (:3000)  →  apps/api (:3001)  →  PostgreSQL 16
                  |                    |
                  |                    →  Redis (timer, BullMQ, pub/sub)
packages/contracts · @kloqra/ui · @kloqra/web-shared
```

NestJS

Prisma

Next.js 15

Zod

Zustand

Tailwind v4

Socket.IO

SSE

API modules — 17 vertical slices

Each module: domain/ → application/ → infrastructure/ → interface/http/

auth workspace users projects tasks categories timelogs timer timesheets billing reporting presence export
notifications assistant jira

200+ routes cataloged in docs/api/ROUTES.md with controller links

Shipped today — production baseline

Domain	Highlights
Core loop	Timer (pause/resume), timesheet, time tracker, tasks, projects, categories + bulk import
Workflow	Per-project approval, submissions, amendments, locked entries
Admin ops	Dashboard widgets, team live (SSE), billing, exports (async jobs, invoice)
Data platform	PostgreSQL range partitioning on time_logs
Realtime	Socket.IO + scoped refetch (submissions, timesheet, projects, tasks)
Quality	CI 4-stage · 19 Playwright e2e · Sentry · deploy runbooks

Maturity: ~7.5/10 — pilot-ready; H0 hardening before broad public launch

Member app — capture time

Demo path

- **/timer** — start on Client Portal Redesign / UX research
- **/timesheet** — week grid, locked approved entries
- **/submissions** — submit week → pending approval

Also shipped

- Time tracker, dashboard widgets
- Export my data, AI assistant
- Jira issue linking
- Onboarding tour

Login: member@kloqra.dev · Acme Corporation workspace

Product app — operate the org

Demo path

- **/projects** — budget burn, approval, team
- **/approvals** — pending, amendments, reminders
- **/dashboard** — widgets, heatmaps
- **/exports** — wizard, jobs, invoice PDF

Login: admin@kloqra.dev

Power features

- Global search (Cmd+K)
- Team live presence (SSE)
- Billing rates & summary
- Impersonation (read-only member view)
- Public reporting API keys

Hero moment — realtime live sync

Two browsers: member on Submissions · product approves OR assigns task

```
createInApp → Redis PUBLISH notifications:user:{userId}
             → NotificationsGateway (Socket.IO /notifications)
             → notification-socket-manager (browser)
             → bell + invalidateWorkspaceData(scopes)
             → pages REST-refetch (source of truth)
```

Scopes: submissions · timesheet · projects · tasks · pending_approvals

Safety net: 60s poll when socket down · reconnect broad catch-up

Dual realtime transports

Use case	Transport	Path
User notifications	Socket.IO	/notifications
Admin presence	SSE	/presence/stream
Multi-replica fan-out	Redis pub/sub	Railway API pods

Intentionally deferred

Workspace timezone broadcast · cosmetic project rename — refresh on navigate is sufficient

Scale — database partitioning

- **time_logs** — monthly partitions on `start_time`
- **time_log_audit_events** — yearly partitions on `created_at`
- Compound PKs for PostgreSQL partition rules
- Cursor pagination: `id:startTime` — zero lookup

Partition pruning on timesheet queries · O(1) compliance data drop

CI / CD pipeline

Stage	Validates
quality	Prettier, ESLint, TypeScript, Turbo build, bundle budget
unit	Vitest + coverage (api, contracts, ui, web-shared)
integration	Postgres 16 + Redis 7, migrate, seed, Supertest
e2e	Playwright — 19 specs (client + admin)
deploy	migrate → Railway API → Vercel → smoke checks

Engineering discipline

docs/specs/ → packages/contracts → failing tests → API → FE → CI gate

- **14 feature specs** in docs/specs/
- **Pre-commit:** tests required on staged feature changes
- **Swagger:** localhost:3001/api/docs
- **Test hub:** pnpm test:dashboard → coverage + Playwright reports

Agentic-assisted development

Cursor multi-agent factory — humans orchestrate, agents implement, CI judges

Rules (.cursor/rules/)

- master-orchestrator — contract-first
- role-be / role-fe / role-qa — directory bounds
- contracts-gate, testing-tdd

Skills + state

- chronomint-feature-delivery
- chronomint-test-delivery
- TASK_BOARD.json (P1/P2 done)
- MIP handoff protocol

Case study — WebSocket notifications

Step	Deliverable
Problem	Product approves; member UI stale up to 60s
Contracts	notification-realtime.ts, scopes, templates
API	Gateway + Redis publisher on createInApp
FE	socket-manager + workspace-data-sync + tasks scope
Quality	Unit specs across api, web-shared, client, contracts

Roadmap horizons

H0 Launch hardening



H1 Workflow excellence



H2 Finance & client value



H3 Scale



H4 Platform & AI

Don't start H3 until H0 exit criteria are met.

Full plan: docs/architecture/KLOQRA_FUTURE_PLAN.md

H0 — Launch hardening (0–6 weeks)

Initiative	Deliverable
Production gate	Rate limits on auth, export, heavy list endpoints
Realtime ops	WSS + Redis runbook; manual verify matrix
Alerting	Sentry DSN; error-rate alerts; /health uptime
Merge train	dev → main with full CI + deploy smoke

Exit: 2+ pilot workspaces · zero P0 bugs for 2 weeks · realtime on prod

Epic: P3-01

H1 — Workflow excellence (6–14 weeks)

P0 Product	P1 Product	Engineering
Budget burn-down widget Budget / idle notifications	Project detail hub Member quick actions Personal goals	Audit log v1 Realtime e2e tests Workspace settings Zod SSOT

Epics: P3-02 · P3-03 · P3-04

H2 — Finance & client value (3–6 months)

- Utilization report — member × week vs expected hours
- Invoice PDF v1 — draft from billable export
- Scheduled export email — SMTP for ExportSchedule
- Period compare — month-over-month delta
- Multi-currency / tax lines — research spike

Epics: P3-05 · P3-06 · **Not building:** QuickBooks replacement

H3 — Scale & external users (6–12 months)

- **Client portal** — external read-only login for client orgs
- **PWA / mobile timer** — installable; offline queue (phase 2)
- **Cross-workspace export** — agency multi-workspace
- **Read replicas** — reporting off primary
- **Email / push reminders** — missing-time digests

Epics: P3-08 · P3-09

H4 — Platform & intelligence (12+ months)

Theme	Ideas
Integrations	Outbound webhooks; deeper Jira sync; Zapier/Make
AI	Smart categorization worker; assistant confirm-before-write
Marketplace	Public API partners

Epic: P3-10 · Deferred until H1–H2 stable

Success metrics

Metric	H0 target	H1 target
Pilot workspaces	3+	10+
Weekly active loggers	—	70% of seats
Submit → approve median	—	<24h
P0 incidents / month	0	≤1
E2E pass on main	100%	100%
Socket connected %	80%	90%

Risks & out of scope

Risks

- Realtime silent failure → runbook + poll safety net
- Export overload → BullMQ + row limits
- Client portal creep → auth spike first
- Agent drift → TASK_BOARD + CI

Out of scope

- AI categorization before audit log
- ERP / accounting replacement
- Cross-workspace billing
- Idle extension before PWA
- Live sync for cosmetic settings

40-minute demo run-of-show

Min	Block
0–3	Hook & positioning
3–8	Architecture + completeness
8–16	Member app + realtime hero moment
16–24	Product app (approvals, exports, dashboard)
24–30	Technical deep dive (contracts, partitioning)
30–35	CI, tests, documentation
35–40	Agentic development + close

Script: docs/user-guides/demo-40min-script.md

Thank you

Kloqra — built for agencies, shipped with discipline

[docs/architecture/KLOQRA_FUTURE_PLAN.md](#)

[docs/user-guides/demo-40min-script.md](#)

[docs/api/ROUTES.md](#)